

Question 10

Correct

Mark 1.00 out of 1.00

Flag question

Indicate for each code sequence given below, supposed to be run by a shell, the handled command line.

1. `cat < mouse`

```
pid = fork();
if ( pid > 0 ) {
    wait(NULL);
} else {
    fd = open("mouse", O_RDONLY);
    dup2(fd, 0);
    close(fd);
    execlp("cat", "cat", NULL);
    perror("execl has not succeeded");
}
```

2. `cat cat > mouse`

```
pid = fork();
if ( pid > 0 ) {
    wait(NULL);
} else {
    fd = open("mouse", O_WRONLY);
    dup2(fd, 1);
    close(fd);
    execlp("cat", "cat", "cat", NULL);
    perror("execl has not succeeded");
}
```

3. `cat cat > mouse &`

```
pid = fork();
if ( pid > 0 ) {
} else {
    fd = open("mouse", O_WRONLY);
    dup2(fd, 1);
```



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LEGION

Question 9

Correct

Mark 1.00 out of 1.00

Flag question

An operating system is closely tied to (i.e. dependent by) the hardware it runs on.

Select one:

☒ True ✓

☐ False

Question 10

Correct

Mark 1.00 out of 1.00

Flag question

The operating system usually allows user applications directly access and control system's resources.

Select one:

☐ True

☒ False ✓

Question 11

Correct

Mark 1.00 out of 1.00

Flag question

Match the following command lines with their corresponding effect.

cat test > file.err

STDOUT redirection

✓

cat test 2> results

STDERR redirection

✓

cat < test

STDIN redirection

✓

cat < file1 > file2

STDIN and STDOUT redirection

✓

cat test 1>f1 2>f2

STDOUT and STDERR redirection

✓

Your answer is correct.

Question 18

Correct

Mark 1.00 out of 1.00

Flag question

You are given the following code, which creates a file. Supposing the file is created successfully, which of the following actions can be performed successfully on the created file.

```
int fd = create("test", 0651);
```

Select one or more:

- ☒ a. owner can read the file ✓
- ☒ b. owner can write the file ✓
- ☐ c. owner can execute the file
- ☒ d. users in group with owner can read the file ✓
- ☐ e. users in group with owner can write the file
- ☒ f. users in group with owner can execute the file ✓
- ☐ g. other users can read the file
- ☐ h. other users can write the file
- ☒ i. other users can execute the file ✓

Your answer is correct.

Question 19

Correct

Mark 1.00 out of 1.00

Flag question

Which of the following could be valid process states during a process lifetime?

Select one or more:

- ☒ a. running ✓
- ☒ b. ready ✓
- ☒ c. blocked ✓

Flag question

```
int count = 0;

void* increment(void* arg)
{
    count++;
}

int main(int argc, char **argv)
{
    pthread_t th[N];

    for (i = 0; i < N; i++) {
        pthread_create(&th[i], NULL, increment, NULL);
    }

    for (i = 0; i < N; i++) {
        pthread_join(th[i], NULL);
    }
}
```

Select one or more:

- ☒ a. a lock ✓
- ☒ b. a semaphore initialized with 1 ✓
- ☐ c. a semaphore initialized with 10
- ☐ d. a semaphore initialized with 0
- ☐ e. no synchronization is needed, while the instruction is the critical region is atomic

Your answer is correct.

Question 25

Correct

Mark 1.00 out of 1

Which is the size in bytes of a process virtual address space in a system using 4 bits for representing a virtual memory address? You must give a number, not a mathematical formula.

Question 7

Correct

Mark 1.00 out of 1.00

Flag question

You are given a C program, whose code is partially illustrated below. Its purpose is to display the contents of the two files "book_1.pdf" and "book_2.txt" on the screen in a human understandable way. File extensions does not necessarily indicate correctly the file type and it is supposed the program makes the needed checkings before executing the code given below.

```
int fd;
char c;
unsigned int offset, size;

fd = open("book_1.pdf", O_RDONLY);
printf("book_1: ");
while (read(fd, &c, 1) == 1)
    write(1, &c, 1);
close(fd);
printf("\n");

fd = open("book_2.txt", O_RDONLY);
read(fd, &offset, sizeof(offset));
read(fd, &size, sizeof(size));
lseek(fd, offset, SEEK_SET);
printf("book_2: ");
for(int i = 0; i < size && read(fd, &c, 1) == 1; i++)
    write(1, &c, 1);
close(fd);
printf("\n");
```

Supposing that the output of the, when run, is:

book_1: Hello world!

book_2: Here we are during the OS exam! Good luck!

In that case:

1. file "book_1.pdf" is a file

2. file "book_2.txt" is a file



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LEGION

Marks 25.53/27.00

Grade 9.46 out of 10.00 (95%)

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

An *operating system* is a special **software**, whose role is to **manage** computer's resources and provide the **users** a convenient and safe environment to **run** their **programs** in.

Question 2

Correct

Mark 1.00 out of 1.00

Flag question

From a *logical perspective*, the OS position in a computing system's configuration is **above** the hardware and **below** the user applications.

Question 3

Correct

Mark 1.00 out of 1.00

Flag question

The *two roles* an operating system plays in a computing system are to

1. **provide** an **abstract** interface of that system and
2. **manage** that system's **resources**.

Question 4

Match the following software types with the CPU execution mode they usually run in.

<https://moodle3.cs.utcluj.ro/mod/quiz/review.php?attempt=52735&cmid=18212#question-61279-15>

9	10	11
✓	✓	✓
17	18	19
✓	✓	✓
25	26	27
✓	✗	✓

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Marks 25.53/27.00
Grade 9.46 out of 10.00 (95%)

Question 1

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<https://moodle3.cs.utcluj.ro/mod/quiz/review.php?attempt=52735&cmid=18212#question-61279-15>

9	10	11	12
✓	✓	✓	✓
17	18	19	20
✓	✓	✓	✓
25	26	27	
✓	✗	✓	

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Started on Wednesday, 17 June 2020, 11:45 AM
State Finished
Completed on Wednesday, 17 June 2020, 12:20 PM
Time taken 35 mins 1 sec
Marks 11.15/14.00
Grade 7.96 out of 10.00 (80%)

Question 1

Correct

Mark 1.00 out of 1.00

 Flag question

How many switches from user to kernel mode will trigger the execution of the following code, supposing that the only files existing in its current working directory are "test_2", "test_3", and "test_6" (with read access for everyone) and the size of the opened file, in case it exists, is 5?

```
int x = 2;
char c;

if (x > 0) {
    printf("x = %d\n", x);
}

int fd = open("test_3", O_RDONLY);
if (fd < 0) {
    perror("Cannot open file");
    exit(1);
}

int n = read(fd, &c, 1);
if (n > 0) {
    write(1, &c, 1);
}

exit(0);
```

Answer: 5



```
printf("The value at address %p is: %d\n", p, *p);  
  
exit(0);  
}
```

2. The process running the following program will be terminated **voluntarily** because of

calling the exit() syscall

```
int main(int argc, char **argv)  
{  
    int x = 10;  
    int *p = &x;  
  
    if (argc != 2) {  
        printf("Usage error\n");  
        exit(10);  
    }  
  
    printf("The value at address %p is: %d\n", p, *p);  
  
    exit(0);  
}
```

3. The process running the following program will be terminated **involuntarily** because of

triggering a division-by-zero exception

```
int main(int argc, char **argv)  
{  
    int x = 10;  
    int y = 0;  
  
    printf("x = %d\n", x);  
    printf("x / y = %d\n", x / y);  
  
    exit(0);  
}
```



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Your answer is partially correct.

Question 6

Partially correct

Mark 0.86 out of 1.00

Flag question

The following code contains a **path traversal** vulnerability, which is triggered by the instruction at line

10, which **opens a file**, located **outside** the intended **directory**.

```
1: #define MAX 1024
2:
3: int main (int argc, char **argv)
4: {
5:     char path[MAX], line[MAX];
6:     int fd, len;
7:
8:     snprintf(path, MAX, "/var/www/os/%s", argv[1]);
9:
10:    fd = open(path, O_RDONLY);
11:    if (fd < 0 ) {
12:        perror("Cannot open file");
13:        exit(1);
14:    }
15:
16:    int n = read(fd, line, MAX);
17:
18:    write(1, line, n);
19: }
```

One fix of the vulnerability consists in inserting the following code

```
if (strstr(argv[1], "..") != NULL) { exit(2); }
```

at line **if (strstr(argv[1], "..") != NULL) { exit(2); }**.



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Question 4

Correct

Mark 1.00 out of 1.00

Flag question

Match the following software types with the CPU execution mode they usually run in.

operating system	kernel mode (i.e. the most privileged mode)	✓
user applications	user mode (i.e. the least privileged mode)	✓

Your answer is correct.

Question 5

Correct

Mark 1.00 out of 1.00

Flag question

A fundamental hardware support an OS uses to protect itself and the hardware from user processes is:

Select one:

- ☒ a. a hierarchy of CPU execution modes with different privileges, the OS running in the most privileged one, while the user applications in the less privileged ones ✓
- ☐ b. different processors used to run the OS and user applications, respectively, the OS running on the boot-strap processor, while the user applications on the application processors
- ☐ c. different memory chips used to store the OS and the user applications, respectively, the OS being loaded on the fastest one, while the user applications on the slower ones
- ☐ d. different disks used to store the OS and the user applications, respectively, the OS on the encrypted disk, while the user applications on the unencrypted ones

Your answer is correct.

2. file "book_2.txt" is a **binary** file

Question 8

Correct

Mark 1.00 out of 1.00

Flag question

You are given the following code, which creates a file. Supposing the file is created successfully, which of the following actions can be performed successfully on the created file.

```
umask("0037");  
int fd = create("test", 0651);
```

Select one or more:

- ☒ a. owner can read the file ✓
- ☒ b. owner can write the file ✓
- ☐ c. owner can execute the file
- ☒ d. users in group with owner can read the file ✓
- ☐ e. users in group with owner can write the file
- ☐ f. users in group with owner can execute the file
- ☐ g. other users can read the file
- ☐ h. other users can write the file
- ☐ i. other users can execute the file

Your answer is correct.

Question 9

Correct

Mark 1.00 out of 1.00

You are given the C program below, supposed to be run on a Linux OS. You must indicate (as a comment) for each `printf()` instruction the number of times the corresponding message will be displayed, supposing all instructions (and called functions) will be executed successfully.



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Flag question

- ☒ a. running ✓
- ☒ b. ready ✓
- ☒ c. blocked ✓
- ☒ d. terminated (like the Linux zombie state) ✓
- ☐ e. jumping
- ☐ f. flattering
- ☐ g. jabbering

Your answer is correct.

Question 20

Partially correct

Mark 0.60 out of 1.00

Flag question

Which are the basic rules a pipe works?

Select one or more:

- ☒ a. bytes are read in the same order they were written (FIFO rule) ✓
- ☒ b. a process trying to read from an empty pipe is blocked, if there are processes having the pipe opened for write (blocking-read rule) ✓
- ☒ c. a process trying to write in a full pipe is blocked, if there are processes having the pipe opened for read (blocking-write rule) ✓
- ☐ d. a process trying to read from an empty pipe is not blocked, if there are no processes having the pipe opened for write (blocking-read exception)
- ☐ e. a process trying to write in a pipe for which there is no process having the pipe opened for read, is interrupted (blocking-write exception)

Your answer is partially correct.

You have correctly selected 3.

Question 12

Correct

Mark 1.00 out of 1.00

Flag question

Establish for the following file paths their correct type.

"/home/os/file" is a(n) path

"../os/file" is a(n) path

"file" is a(n) path

"../home/os/file" is a(n) path

"/" is a(n) path

Question 13

Correct

Mark 1.00 out of 1.00

Flag question

Associate the following file names their corresponding format type, based on regular user conventions regarding the file extensions:

1. "sort.c" is a file

2. "index.html" is a file

3. "program.zip" is a file

4. "program.o" is a file

5. "book.pdf" is a file

6. "project.docx" is a file

7. "text.xml" is a file

8. "cmd.exe" is a file

Marks 25.53/27.00
Grade 9.46 out of 10.00 (95%)

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

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Question 2

Correct

Mark 1.00 out of 1.00

Flag question

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Question 3

Correct

Mark 1.00 out of 1.00

Flag question

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9	10	11	12	13	14	15	16
✓	✓	✓	✓	✓	✓	✓	Correct
17	18	19	20	21	22	23	24
✓	✓	✓	✗	✓	✓	✓	✓
25	26	27					
✓	✗	✓					

Show one page at a time

Finish review

Question 21

Correct

Mark 1.00 out of 1.00

Flag question

Which types of pipes do the Linux and Windows operating systems provide?

Select one or more:

- ☒ a. anonymous ✓
- ☒ b. named ✓
- ☐ c. unidirectional
- ☐ d. reader-writer
- ☐ e. triangular

Your answer is correct.

Question 22

Correct

Mark 1.00 out of 1.00

Flag question

How many threads could run in the same time in the critical region of the function `aeroplane()` in the code below?

```
sem_t airport;

void* aeroplane(void* arg)
{
    sem_wait(&airport);

    // use a runway to take off or land

    sem_post(&airport);
}

int main(int argc, char **argv)
{
    pthread_t th[32];

    sem_init(&airport, 0, 6);

    for (i = 0; i < 32; i++) {
        pthread_create(&th[i], NULL, aeroplane, NULL);
    }
}
```

Question 5

Partially correct

Mark 0.78 out of 1.00

Flag question

You are given the following C code. You have to indicate (in given comments), which corresponding called functions execute successfully and whether they display something on the screen, in the file or has no effect at all.

```
int fd = open("file", O_WRONLY);
```

```
printf("1\n"); // executes with SUCCESS and displays on the screen
```

```
write(fd, "a", 1); // executes with SUCCESS and writes in the file
```

```
write(1, "a", 1); // executes with SUCCESS and displays on the screen
```

```
close(1);
```

```
printf("2\n"); // executes with FAILURE and has no effect
```

```
write(1, "a", 1); // executes with FAILURE and has no effect
```

```
write(2, "aa", 2); // executes with SUCCESS and has no effect
```

```
dup2(fd, 1);
```

```
printf("3\n"); // executes with FAILURE and has no effect
```

```
write(1, "a", 1); // executes with SUCCESS and writes in the file
```

```
write(2, "aa", 2); // executes with SUCCESS and writes in the file
```

Your answer is partially correct.

Question 6



Search



LEGION

- ☒ h. pipe(fd); ✓
- ☒ i. stat("test", &inode); ✓
- ☐ j. while (x > 0);

Your answer is partially correct.
You have correctly selected 5.

Question 8

Partially correct
Mark 0.80 out of 1.00

Flag question

Usually, an operating system is expected to provide the following properties:

Select one or more:

- ☒ a. provide a powerful and general functionality ✓
- ☒ b. protect itself from user processes and user processes by one another ✓
- ☒ c. be invisible in terms of amount of resources it consumes for itself and the performance overhead it implies ✓
- ☐ d. provide the user the possibility to solve very complex mathematical equations
- ☒ e. protect a computer it runs from physical damage of a possible attacker ✗

Your answer is partially correct.
You have selected too many options.

Question 9

Correct
Mark 1.00 out of 1.00

An operating system is closely tied to (i.e. dependent by) the hardware it runs on.

Select one:

```
execlp("cat", "cat", "cat", NULL);
perror("execl has not succeeded");
}
```

3. cat cat > mouse &

```
pid = fork();
if ( pid > 0 ) {
} else {
    fd = open("mouse", O_WRONLY);
    dup2(fd, 1);
    close(fd);
    execlp("cat", "cat", "cat", NULL);
    perror("execl has not succeeded");
}
```

Question 11

Partially correct
Mark 0.63 out of 1.00

Flag question

Indicate for each program given below the way the process running it will be terminated, i.e. *voluntarily* (on process' explicit request) or *involuntarily* (forcefully by OS) and the termination reason.

1. The process running the following program will be terminated involuntarily because of

calling the printf() syscall

```
int main(int argc, char **argv)
{
    int *p = NULL;

    printf("The value at address %p is: %d\n", p, *p);

    exit(0);
}
```

Answer: 5

Question 2

Correct

Mark 1.00 out of 1.00

Flag question

Match the standard channel redirections from the following command lines with the system calls that are affected by the corresponding redirection.

cat test > file.err

write(1, buf, strlen(buf)); ✓

cat test 2> results

write(2, buf, strlen(buf)); ✓

cat < test

read(0, &c, 1); ✓

Your answer is correct.

Question 3

Incorrect

Mark 0.00 out of 1.00

Flag question

Which is the **maximum number of text lines** the following code run on Linux could read from a text file, supposing the file exists, could be opened for reading, but its length and contents could be variable (i.e. not known in advanced), though the file's length is always **greater than** the value of "MAX"?

```
#define MAX 648

char lines[MAX];

int fd = open("TEST.txt", O_RDONLY);
if (fd < 0) {
    perror("Cannot open file");
    exit(1);
}

int n = read(fd, lines, MAX);

int no_of_lines = 0;
for(int i = 0; i < n; i++) {
    if (lines[i] == '\n') {
        no_of_lines++;
    }
}
```

```
pthread_t th[32];  
  
sem_init(&airport, 0, 6);  
  
for (i = 0; i < 32; i++) {  
    pthread_create(&th[i], NULL, aeroplane, NULL);  
}  
  
for (i = 0; i < 32; i++) {  
    pthread_join(th[i], NULL);  
}  
}
```

Answer: 6 ✓

Question 23

Correct

Mark 1.00 out of 1.00

Flag question

A lock is a **synchronization** mechanism that allows a **single** thread be at one moment in the **critical region** protected by that lock. It provides the **lock()** and **unlock()** primitives.

Question 24

Correct

Mark 1.00 out of 1.00

Flag question

You are given the following code. Which kind of synchronization mechanism could you use to provide correct protection (i.e. synchronization) of the critical region in function `increment()`?

```
#define N 100  
  
int count = 0;  
  
void* increment(void* arg)  
{  
    count++;  
}
```

Question 9

Correct

Mark 1.00 out of 1.00

Flag question

You are given the C program below, supposed to be run on a Linux OS. You must indicate (as a comment) for each `printf()` instruction the number of times the corresponding message will be displayed, supposing all instructions (and called functions) will be executed successfully.

```
int main(int argc, char **argv)
{
    printf("Life in wonderful!\n");           // displayed 1 times

    int pid_1 = fork();

    printf("Life is wonderful!\n");           // displayed 2 times

    int pid_2 = fork();

    printf("Life is wonderful!\n");           // displayed 4 times

    if (pid_1 == 0 || pid_2 == 0) {
        printf("Life is wonderful!\n");       // displayed 3 times
        execlp("ps", "ps", 0);
        printf("Life is wonderful!\n");       // displayed 0 times
        exit(0);
    }

    printf("Life is wonderful!\n");           // displayed 1 times

    fork();

    printf("Life is really wonderful!\n");     // displayed 2 times
}
```



Search



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Your answer is correct.

Question 16

Correct

Mark 1.00 out of 1.00

Flag question

Match the following concepts to their correct definition.

file	the basic storage unit on a file system	✓
directory	a fundamental data organization mechanism	✓
process	a running a program in a "virtual computer"	✓
thread	a particular execution path in a running program	✓

Your answer is correct.

Question 17

Correct

Mark 1.00 out of 1.00

Flag question

The two main components of a file in any file systems are:

Select one or more:

- ☒ a. the file's data (i.e. contents) ✓
- ☒ b. the file's metadata structure ✓
- ☐ c. file's owner
- ☐ d. file's parent directory
- ☐ e. file's access permissions

Your answer is correct.

question

- ☒ b. segmentation ✓
- ☒ c. paging ✗
- ☐ d. swapping
- ☐ e. best fit

Your answer is partially correct.

You have correctly selected 1.

Question 27

Correct

Mark 1.00 out of 1.00

Flag question

Which of the following memory allocation strategies suffers by external fragmentation?

Select one or more:

- ☒ a. single contiguous area for a process ✓
- ☒ b. segmentation ✓
- ☐ c. paging
- ☐ d. swapping
- ☐ e. best fit

Your answer is correct.

Finish review

6. "project.docx" is a file

7. "text.xml" is a file

8. "cmd.exe" is a file

Question 14

Correct

Mark 1.00 out of 1.00

Flag question

From the user perspective, a *text file* is a file in which every is a code of a printable and also contains special codes for marking the of a text .

Question 15

Correct

Mark 1.00 out of 1.00

Flag question

Match the following file types to their *access method* the OS usually imposes to user applications over accesses to their contents.

regular files ✓

directories ✓

pipes ✓

Your answer is correct.

Question 16

Match the following concepts to their correct definition.

Question 6

Correct

Mark 1.00 out of 1.00

Flag question

Which are the two mechanisms used to switch from user to kernel mode and give the OS the possibility to run and handle accesses to the computer's resources?

Select one or more:

- ☒ a. system calls, triggered through dedicated instructions ✓
- ☒ b. exceptions and interrupts ✓
- ☐ c. backdoor instructions
- ☐ d. there is no need, while the OS always run an a computer processors

Your answer is correct.

Question 7

Partially correct

Mark 0.83 out of 1.00

Flag question

Which of the following instructions in a C program run on a Linux OS will normally trigger a system call?

Select one or more:

- ☒ a. `read(fd, buf, 10);` ✓
- ☐ b. `printf("Hello world!\n");`
- ☒ c. `write(1, &c, 1);` ✓
- ☐ d. `i++;`
- ☐ e. `power(2,5);`
- ☐ f. `log10(100);`
- ☒ g. `fork();` ✓
- ☒ h. `pipe(fd);` ✓
- ☒ i. `stat("test", &inode);` ✓
- ☐ j. `while (x > 0);`

Answer: 0 ✗

Question 4

Partially correct

Mark 0.50 out of 1.00

Flag question

How many subdirectories will be counted by the following program in the directory `"/tmp/TEST_DIR"`, also created by the given program? It is supposed that the `"/tmp/TEST_DIR"` does not exist before the program is run, and both its creation and those of its subdirectories succeed.

```
int main(int argc, char **argv)
{
    DIR *dir_handle;
    struct dirent *dir_elem;
    struct stat inode;
    char path[1024];
    int dir_count = 0;

    mkdir("/tmp/TEST_DIR");

    for (int i = 0; i < 775; i++) {
        snprintf(path, 1024, "/tmp/TEST_DIR/DIR%03d", i);
        mkdir(path);
    }

    dir_handle = opendir("/tmp/TEST_DIR");

    while ((dir_elem = readdir(dir_handle)) != NULL) {
        snprintf(path, 1024, "/tmp/TEST_DIR/%s", dir_elem->d_name);
        lstat(path, &inode);
        if (S_ISDIR(inode.st_mode))
            dir_count++;
    }

    printf("dir_count = %d\n", dir_count);
}
```

Answer: 775 ✓

You missed the `".."` and `"."` directories automatically created in every new directory.



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Your answer is correct.

Question 3

Incorrect

Mark 0.00 out of 1.00

Flag question

Which is the **maximum number of text lines** the following code run on Linux could read from a text file, supposing the file exists, could be opened for reading, but its length and contents could be variable (i.e. not known in advanced), though the file's length is always greater than the value of "MAX"?

```
#define MAX 648

char lines[MAX];

int fd = open("TEST.txt", O_RDONLY);
if (fd < 0) {
    perror("Cannot open file");
    exit(1);
}

int n = read(fd, lines, MAX);

int no_of_lines = 0;
for(int i = 0; i < n; i++) {
    if (lines[i] == '\n') {
        no_of_lines++;
    }
}

printf("no of lines = %d\n", no_of_lines);
```

Answer: 0



Question 4

Partially correct

Mark 0.50 out of 1.00

Flag question

How many subdirectories will be counted by the following program in the directory `"/tmp/TEST_DIR"`, also created by the given program? It is supposed that the `"/tmp/TEST_DIR"` **does not exist before the program is run**, and both its creation and those of its subdirectories succeed.

```
int main(int argc, char **argv)
{
    DIR *dir_handle;
    struct dirent *dir_elem;
    struct stat inode;
```



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Your answer is partially correct.

Question 13

Partially correct

Mark 0.50 out of 1.00

Flag question

You are given the following C program, that will be run on an OS using paging for memory management, with pages of 4 KiB (i.e. 4096 bytes). You have to indicate (in given comments), which virtual page (i.e. its number) each instruction will make reference to. We ignore if that page is mapped or not, being or not accessible, respectively.

```
int main()
{
    char *p = 4096;

    p[0] = 0;      // refers virtual page no 0
    p[10] = 1;     // refers virtual page no 1
    p[4096] = 2;   // refers virtual page no 2
    p[5000] = 3;   // refers virtual page no 2

    int *pi = 40960;

    pi[0] = 0;     // refers virtual page no 0
    pi[4096] = 10; // refers virtual page no 1
}
```

Your answer is partially correct.

Question 14

Correct

Supposing that a computer has 56 (logical) processors, how many threads could be simultaneously in the *running* state?



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☐ e. no synchronization is needed, while the instruction is the critical region is atomic

Your answer is correct.

Question 25

Correct

Mark 1.00 out of 1.00

Flag question

Which is the size in bytes of a process virtual address space in a system using 4 bits for representing a virtual memory address? You must give a number, not a mathematical formula.

Answer: 16 ✓

Question 26

Partially correct

Mark 0.30 out of 1.00

Flag question

Which memory management strategies use base and limit registers for translating virtual addresses to physical addresses?

Select one or more:

- ☐ a. single contiguous area for a process
- ☒ b. segmentation ✓
- ☒ c. paging ✗
- ☐ d. swapping
- ☐ e. best fit

Your answer is partially correct.

You have correctly selected 1.

```
sem_wait(&is_boy);

printf("Girl %d met the boy %d\n", met_girl, met_boy);

sem_post(&mutex_2);
}

int main(int argc, char **argv)
{
    pthread_t th[100];
    int i, th_arg[100];

    sem_init(&mutex_1, 0, 1);
    sem_init(&mutex_2, 0, 1);
    sem_init(&is_boy, 0, 0);
    sem_init(&is_girl, 0, 0);

    for (i = 0; i < 100; i++) {
        th_id[i] = i;
        if (i % 2 == 0)
            pthread_create(&th[i], NULL, boy, &th_arg[i]);
        else
            pthread_create(&th[i], NULL, girl, &th_arg[i]);
    }

    for (i = 0; i < 100; i++) {
        pthread_join(th[i], NULL);
    }
}
```

Your answer is partially correct.



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```
printf("x = %d\n", x);
printf("x / y = %d\n", x / y);

exit(0);
}
```

4. The process running the following program will be terminated **involuntarily** because of

triggering a page-fault exception

```
int main(int argc, char **argv)
{
    printf("Hello world!\n");
}
```

Your answer is partially correct.

Question 12

Partially correct

Mark 0.89 out of 1.00

Flag question

You are given the following program creating 100 threads, half of them executing concurrently the function "boy()", while the other half executing concurrently the function "girl()". You are required to complete the gaps such that to provide the correct synchronization, based on the *fair rendezvous pattern*.

```
int met_boy;
int met_girl;

sem_t mutex_1, mutex_2, is_boy, is_girl;

void boy(void *arg)
{
    int th_id = *(int*)arg;

    sem_wait(&mutex_1);

    met_boy = th_id;
```



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```
p[0] = 0;      // refers virtual page no 0
p[10] = 1;     // refers virtual page no 1
p[4096] = 2;   // refers virtual page no 2
p[5000] = 3;   // refers virtual page no 2

int *pi = 40960;
pi[0] = 0;     // refers virtual page no 0
pi[4096] = 10; // refers virtual page no 1
}
```

Your answer is partially correct.

Question 14

Correct

Mark 1.00 out of 1.00

Flag question

Supposing that a computer has 56 (logical) processors, how many threads could be simultaneously in the *running* state?

Answer: 56



Finish review



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```
{
    int th_id = *(int*)arg;

    sem_wait(&mutex_1);

    met_boy = th_id;

    sem_post(&is_boy);

    sem_wait(&is_girl);

    printf("Boy %d met the girl %d\n", met_boy, met_girl);

    sem_post(&mutex_1);
}

void girl(void *arg)
{
    int th_id = *(int*)arg;

    sem_wait(&mutex_2);

    met_girl = th_id;

    sem_post(&mutex_2);

    sem_wait(&is_boy);

    printf("Girl %d met the boy %d\n", met_girl, met_boy);

    sem_post(&mutex_2);
}

int main(int argc, char **argv)
{
    pthread_t th[100];
    int i, th_arg[100];
```



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